

Sadhana Jamkatel

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EDUCATION

North Central College, Naperville, IL **GPA 3.85/4** *Expected: May 2027*

Major: Bachelor of Science in Computer Science, **Minor:** Economics, **Concentration:** Ethical Leadership

Relevant Coursework: Data/Discrete Structures, Computer Organization/Design, Software Development(OOP), Data Mining, Database System, Algorithm & Design, Computer Network & Security

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, Java

Web & Frameworks: React.js, Node.js, Flask, Next.js, HTML/CSS

Cloud & DevOps: AWS (S3, EC2, Lambda, CloudFront, CloudWatch), Docker, GitHub Actions, Linux/Unix, CI/CD

Data & ML: PostgreSQL, MongoDB, scikit-learn, pandas, NumPy, LangChain

WORK EXPERIENCE

DevOps Engineer Intern| Intern of the Season 2025

June 2025-Aug 2025

Aeropay

- Reduced documentation search time from 30 minutes to 30 seconds by building a full-stack internal documentation site using Docusaurus (MDX, custom charts, integrated search), serving 30+ engineers daily and hosted on GitHub Pages with automated CI/CD via GitHub Actions
- Secured and deployed the site using AWS S3 + CloudFront with JumpCloud SSO access controls and AWS CloudWatch telemetry for performance monitoring, ensuring 99.9% uptime for DevOps and engineering teams
- Reduced developer setup time by ~70% by designing a CLI tool in TypeScript that automated authentication, role-based access, and token generation for AWS/database connections
- Automated CLI release documentation using OCLIF and GitHub Actions, publishing updated docs to a live site with every release and eliminating manual updates

High-Performance Computing (HPC) Bootcamp Participant

August 2025

Argonne National Laboratory

- Built an interactive HPC chatbot for scientific Q&A; delivered RAG and non-RAG versions (LangChain , Huggingface, Gemini), compared accuracy and latency, and documented trade-offs.
- Gained hands-on exposure to Argonne's supercomputing ecosystem (incl. Aurora) and HPC workflows used in scientific computing.
- Completed workshops in data visualization and machine learning for science; practiced turning datasets into clear, usable insights.

PROJECTS

Predictive Maintenance ML System for Fleet Engine Monitoring

Sep 2025

- Reduced engine downtime by 95% by building a predictive maintenance system using Random Forest to analyze 21 sensor readings from NASA Turbofan data and detect at-risk engines before failure.
- Designed React dashboard with real-time alerts and risk visualization to monitor 100+ engines and notify maintenance teams of critical issues.
- Achieved 96% accuracy with 4.28% false alarm rate, delivering \$750K+ projected annual cost savings through optimized hyperparameter tuning and feature engineering.

Medical Recommendation System

Jan 2025 - Feb 2025

- Achieved 92% disease prediction accuracy by training and comparing SVC, Random Forest, and Gradient Boosting models on 10,000+ symptom-treatment pairs, enabling faster preliminary diagnosis for healthcare providers.
- Built a full-stack web application using Flask for RESTful API backend and responsive frontend, integrating Google Search API to deliver real-time treatment suggestions, reducing manual research time from 5+ mins to under 1 min through automated API integration

Twitter Clone | Full Stack Development | Web Application Development

May 2024- Dec 2024

- Developed a full-stack Twitter clone supporting user authentication, tweet posting, and social interactions (following/unfollowing), deployed and tested with 15+ users using React, Vite, Node.js, Express.js, MongoDB, and Cloudinary.
- Designed and implemented RESTful APIs with JWT, bcrypt, and custom middleware, applying software design best practices for scalability and security, and tested using relational database concepts to ensure reliable data handling.

ON CAMPUS INVOLVEMENT

Resident Assistant | North Central College | RA of the Year 2025

May 2024 – Present

- Increased engagement by 80% by organizing 15+ events per semester and providing mentorship to 100+ residents

Vice President | Society of Hispanic Professional Engineers (SHPE) | North Central College

May 2023 – Present

- Secured conference funding and led 10+ career development workshops for 20+ engineering students while overseeing executive board operations and coordinating semester programming

President | Computer Science Club | North Central College

May 2023 – Present

- Grew club to 25+ members by organizing 10+ technical workshops with industry professionals, leading collaborative projects, and fostering peer mentorship to increase retention and hands-on learning.

PROFESSIONAL AFFILIATIONS

Color Stack, RTC, SHPE, NSBE, AnitaB.org, Girls Who Code, ACM, Women in Technology, Codepath, Code2040